WON JUN LEE

Ph.D. Candidate in Electrical and Computer Engineering, Purdue University

Profile

Ph.D. candidate in **wireless network scheduling and resource allocation for 5G/6G systems**. Develops provably optimal, low-complexity scheduling policies (**stochastic optimization / MDP**) for multi-channel communication and multi-sensor estimation, and turns the optimal policy's structure into scalable algorithms and priors for **learning-based network decision-making**.

Contact Details

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Website

Professional Service

Reviewer, IEEE International Symposium on Information Theory (ISIT), *2026*

Awards & Honors

Ross Fellowship, Purdue University, *2021–2025*
Student Travel Grant, IEEE International Symposium on Information Theory (ISIT), *2024*
Dean's List, Yonsei University, *2015–2016, 2019–2020*

Skills

Wireless: *5G/6G Resource Allocation, MAC Scheduling, Age of Information*
Modeling & Optimization: *MDPs, Stochastic Control, Convex Optimization, Probability, Information Theory, Reinforcement Learning (fundamentals)*
Programming: *MATLAB, Python (NumPy, PyTorch), C, Java*

Education

Purdue University — *West Lafayette, IN, U.S.*
Ph.D. Candidate, Electrical and Computer Engineering — *Aug 2021 – Exp. May 2027*
Advisor: Prof. Chih-Chun Wang
GPA: 3.81/4.0

Yonsei University — *Seoul, South Korea*
B.S., Electrical and Electronics Engineering — *Mar 2015 – Feb 2021*
GPA: 3.84/4.0

Research Experience

Prof. Chih-Chun Wang's Lab
Research Assistant, Purdue University — *West Lafayette, IN, U.S.* — *Aug 2021 – Present*

AoI-Optimal Scheduling and Resource Allocation for Multi-Channel Wireless Systems → [C2, J1]

• **Characterized the AoI-optimal scheduling policy** for arbitrary K-channel systems as a semi-Markov decision process on the age state: threshold (water-filling) structure with closed-form solutions, and **reduced policy computation from $O(N^3)$ to $O(N)$**.

• Extended the framework to a **joint latency–cost–quality–freshness optimization** over heterogeneous channels (e.g., cloud vs. edge server selection), with a structured channel-selection algorithm that scales to large systems (journal article in preparation).

Cost-Aware Multi-Sensor Scheduling for Remote Estimation under Correlated Noise → [C3]

• Developed a **cost-aware sensor-scheduling framework** that decides which sensors transmit under communication and sensing budgets, trading estimation error against cost.

• Designed an estimation method that **alleviates the computational burden of Kalman filtering** under correlated noise and remains tractable for high-dimensional systems.

Digital Communications Lab
Undergraduate Research Intern, Yonsei University — *Seoul, South Korea* — *Jan – Jun 2020*

5G Network Performance Benchmarking in Heterogeneous Environments

• Built an Android network-analysis application (**UDP socket programming**) to measure **end-to-end latency and throughput** in heterogeneous 5G/Wi-Fi environments; implemented **UDP hole punching** for real-world connectivity benchmarking.

High Dimensional Signal Processing Lab
Undergraduate Research Intern, Yonsei University — *Seoul, South Korea* — *Jul – Aug 2019*

DNN Classification Logic Analysis → [C1]

• Assisted in analyzing decision boundaries of deep neural network classifiers.

Military Experience

KATUSA (Korean Augmentation To the United States Army) — *Seoul, South Korea*
Sergeant — *Mar 2017 – Dec 2018*

• Information Management Officer (MOS 25B), USFK Public Affairs Office; **led 6 soldiers as squad leader**, managing ∼100 networked workstations and coordinating ROK–U.S. technical operations during **joint exercises** (UFG 2017 / Foal Eagle 2018).

Publications

- **[J1] W.J. Lee** and C.-C. Wang, "Latency-, cost-, and freshness-aware scheduling over heterogeneous multi-channel systems," journal article, *in preparation*.
- **[C3] W.J. Lee** and C.-C. Wang, "Cost-aware sensor scheduling for multi-sensor remote estimation under correlated noise," *submitted to IEEE Int. Conf. on Communications (ICC)*, 2027.
- **[C2] W.J. Lee** and C.-C. Wang, "AoI-optimal scheduling for arbitrary K-channel update-through-queue systems," *in Proc. IEEE Int. Symp. on Information Theory (ISIT)*, 2024, pp. 957–962.
- **[C1]** J. Kim, S. Woo, **W.J. Lee**, D. Kim, and C. Lee, "Analyzing Decision Polygons of DNN-based Classification Methods," *Proc. ICINCO*, 2020, pp. 112–119.